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328655(28)

B.E. (Sixth Semester)
Examination, April/May 2015
(New Course)
(Branch : ET & T)

INFORMATION THEORY AND CODING

Time : Three Hours] [*Maximum Marks : 80*

[*Minimum Pass Marks : 28*

Note : 1. Part (a) of each question is compulsory.
2. All questions carry equal marks.

UNIT-I

Attempt any two parts from (b), (c) and (d).

1. (a) Calculate extended entropy $H(s^2)$ of the source symbol whose probabilities are 0.25, 0.25 0.34 and 0.16.

(b) Explain in details the lossless and lossy compression techniques.

(c) A 2 kHz channel has a signal to noise ratio of 24 dB. Calculate :

(i) Maximum capacity of the channel.

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(ii) Assuming constant transmitting power, calculate the maximum capacity when channel bandwidth is halved

(iii) Assuming constant transmitting power, calculate the maximum capacity when channel bandwidth reduced to a quarter of the original value.

(d) Explain binary symmetrical channel and derive the expression for the mutual information. 7

UNIT-II

Attempt any **Two** parts from (b), (c) and (d).

2. (a) Define block code and syndrome 2

(b) A (6, 3) code is generated according to the generator matrix. The receiver receives Y = 100011. Determine the corresponding data word if the channel is a binary symmetric channel and maximum likelihood decision is used : 7

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

(c) The impulse response of the input top adder output path and input bottom adder output path of convolution encoder is {1, 1, 1} and {1, 0, 1} respectively and message sequence is {1 0 01 1}. Calculate the convolution encoder sequence.

(d) Explain in details the single error correcting hamming codes with suitable example. 7

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UNIT-III

Attempt any **Two** parts from (b), (c) and (d).

3. (a) Calculate run length coding for the sequence Z = {1, 1, 1, 2, 2, A, A, A, B, C, C, C} 2

   (b) Explain BCH code in details with suitable example. 7

   (c) Explain the significance of Reed-Solomon code also explains how it can be used for information technology. 7

   (d) Explain CCITT group-3 and group-4 types image compression with suitable example. 7

UNIT-IV

Attempt any **Two** parts from (b), (c) and (d).

4. (a) Define code rate of convolution code.

   (b) Construct a symmetric (7, 4) cyclic code using the generator polynomial $g(x) = x^3 + x + 1$. What are the error correcting capabilities of this code ? For the received word 1101100, determine the transmitted codeword. 7

   (c) Explain the working of (2, 1, 3) convolution encoder using transform domain approach. 7

   (d) Explain the maximum likelihood decoding and viterbi decoding algorithms of a convolution encoder. 7

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UNIT-V

Attempt any **Two** parts from (b), (c) and (d).

5. (a) Define digital signature. 2

   (b) Explain in details Ungerboeck's TCM Design Rules, also explain TCM decoder. 7

   (c) Consider (3, 1, 2) convolution code with $g(1) = (110)$, $g(2) = (101)$ and $g(3) = (111)$: 7

   (i) Draw the encoder block diagram.

   (ii) Find the generator matrix.

   (iii) Find the code word corresponding to the information sequence (11101) using time domain approach.

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